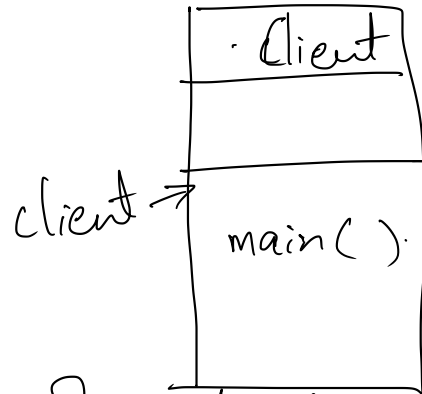
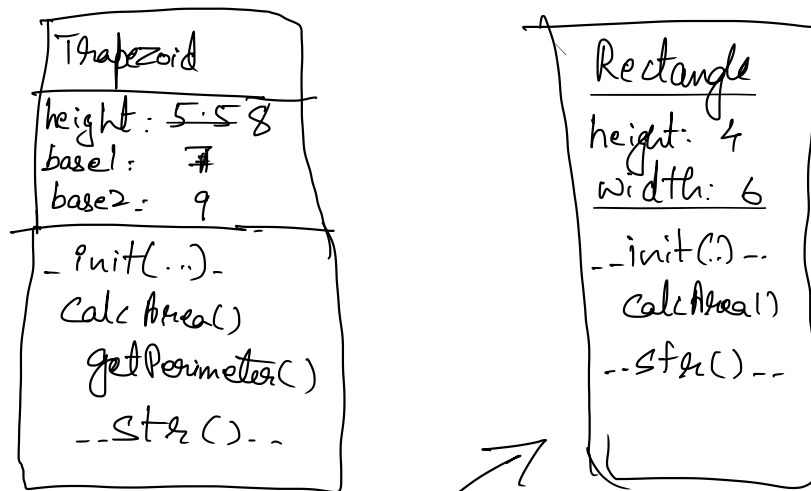


Q3] Code tracing represents how your program gets represented in your computer memory.
Implementation here begins from main() function!



Inside the main() of Client class



shapes = [tr, rect]
For loop: shape: tr → prints
 shape: rect → prints

Output:

①

Trapezoid: height: 5.5, ~~base~~1: 7, base2: 9

$$\text{area: } 44 \Rightarrow (7+9) // 2 = 8 * 5.5 = 44.0$$

Rectangle: height: 4, width: 6

area: None # because of pass None returned

new height: 8

$$\text{new area: } 8 * 6 = 48$$

②

Error when updating tr. -- base1

③

For tr: Trapezoid.calcArea()

For rect: Rectangle.calcArea()

Q6

num_list = [~~5~~, ~~4~~, 0]
i in range(3) = 0, 1, 2

Iteration 1

i = 0

if (num_list[0] % 4)

5 % 4 = 1

if (1) = 1 $\rightarrow$ T

num_list[0] = 2 * func_a(5)

$\downarrow$

sqrt(5) + 1

~~2~~ 3.23 = 6.4

Iteration 2

i = 1

if $\frac{4 \% 4}{(0)} \rightarrow F$ X

elif $\frac{4 \% 5}{(1)} \rightarrow T$ ✓

num_list[1] = func_a(4) + func_b(4, 1)

$\downarrow$

sqrt(4) + 1

= 2 + 1

= 3

$\downarrow$

4 + 2.7⁴

= 4 + 54.5

= 58.5

61.5

$i = 0$ `players[0] = 'Iokic'`

$j = 0$ `favorites[0] ?` ✗

$= 1$ `favorites[1] ?` ✓

2 `favorites[2] ?` ✗

3 `favorites[3]` ✓

$i = 1$ `players[1] = 'SGA'`

$j = 0$ `favorites[0] ?` ✗

$= 1$ `favorites[1] ?` ✗

2 `favorites[2] ?` ✗

3 `favorites[3] ?` ✗

4 `favorites[3]` ✗

$i = 2$ `players[0] = 'Giannis'`

$j = 0$ `favorites[0] ?` ✗

$= 1$ `favorites[1] ?` ✓

2 `favorites[2] ?` ✗

3 `favorites[3]` ✗

$i = 3$ `players[1] = 'Tatum'`

$j = 0$ `favorites[0] ?` ✗

$= 1$ `favorites[1] ?` ✗

2 `favorites[2] ?` ✓

3 `favorites[3]` ✗

$i = 4$ `players[1] = 'Lebron'`

$j = 0$ `favorites[0] ?` ✗

$= 1$ `favorites[1] ?` ✗

2 `favorites[2] ?` ✗

3 `favorites[3]` ✗

Next some prints

Output:

Jokic

Giannis

Lebron

Jokic count: 1

Tatum index: 3

Common players: ('Jokic', 'Tatum')

Number of common players: 2

Jokic is at tuple index 0

Tatum is at tuple index 3

Done